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Iteration 2 (from Hero's Results : $m = 1.7$, $b = 2.1$)

Points (1,3) (3,6)

$$\alpha = 0.1$$

Predicted values:

$$\hat{y}_1 = 1.7(1) + 2.1 = 3.8$$
$$\hat{y}_2 = 1.7(3) + 2.1 = 7.2$$

Errors ($y - \hat{y}$):

$$e_1 = 3 - 3.8 = -0.8$$
$$e_2 = 6 - 7.2 = -1.2$$

Gradients (chain rule) : $\partial E / \partial m = -2/n * \sum (y - \hat{y})x = -2/2 * [(-0.8)(1) + (-1.2)(3)] = -1[-0.8 - 3.6] = 4.4$

$$\partial E / \partial b = -2/n * \sum (y - \hat{y}) = -2/2 [(0.8) + (-1.2)] = -1(-2.0) = 2.0$$

Update :

$$m_{\text{new}} = 1.7 - 0.1(4.4) = 1.7 - 0.44 = 1.26$$
$$b_{\text{new}} = 2.1 - 0.1(2.0) = 2.1 - 0.20 = 1.90$$